MATH 39 — Linear Algebra

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1.1 Linear Equations

Definition A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \ldots, a_n$ and b are constants. The constants $a_1, a_2, \ldots, a_n$ are called the **coefficients** of the equation, and b is called the **constant term**.

Examples of linear equations include:

- $2x + 3y - z = 5$
- $-x_1 + 4x_2 + 0x_3 = 7$
- $0x + 0y + 0z = 0$

Non-examples of linear equations include:

- $x^2 + y = 3$ (because of the x^2 term)
- $xy + z = 4$ (because of the xy term)
- $\sin(x) + y = 1$ (because of the $\sin(x)$ term)
- $x + \sqrt{y} = 2$ (because of the $\sqrt{y}$ term)

Systems of Linear Equations

Definition A **finite** collection of linear equations involving the same set of variables is called a **system of linear equations**.

Example of a system of linear equations:

$$\begin{cases} 2x + 3y - z = 5 \\ -x + 4y + 2z = 7 \\ 3x - y + z = 2 \\ \cdots \end{cases}$$

Definition A sequence of numbers $(s_1, s_2, \ldots, s_n)$ is called a **solution** to the system of linear equations if, when we substitute s_i for x_i in each equation, the equation holds true.

Example Consider the system of equations:

$$\begin{cases} x + y + z = 3 \\ 2x + y + 3z = 1 \end{cases}$$

Is the sequence $(-2, -5, 0)$ a solution to this system? Does $x = y = z = 1$?

Solution To check if $(-2,-5,0)$ is a solution, substitute $x=-2$, $y=-5$, and $z=0$ into both equations:

- First equation: $(-2)+(-5)+0=-7\neq 3$ (does not hold)
- Second equation: $2(-2)+(-5)+3(0)=-4-5=-9\neq 1$ (does not hold)

Thus, $(-2,-5,0)$ is not a solution to the system.
Next, check if $(1,1,1)$ is a solution:

- First equation: $1+1+1=3$ (holds)
- Second equation: $2(1)+1+3(1)=2+1+3=6\neq 1$ (does not hold)

Thus, $(1,1,1)$ is also not a solution to the system.

Solutions to Linear Systems

Definition There are three possible scenarios for the solutions to a system of linear equations:

- **No solution**: The system is inconsistent, meaning there are no values for the variables that satisfy all equations simultaneously.
- **Unique solution**: There is exactly one set of values for the variables that satisfies all equations.
- **Infinitely many solutions**: There are multiple sets of values for the variables that satisfy all equations. (i.e. the same line or scalar multiples of each other)

Systems of linear equations that have at least one solution are called **consistent**, while those with no solutions are called **inconsistent**.

Matricies

Definition A **matrix** is a rectangular array of numbers arranged in rows and columns. The numbers in the matrix are called its **entries** or **elements**. A matrix with m rows and n columns is called an $m\times n$ matrix.

Example of an **augmented** matrix:

$$\begin{cases} 3x_1+2x_2-x_3+x_4=-1 \\ 2x_1-x_3+2x_4=0 \\ 3x_1+x_2+2x_3+5x_4=2 \end{cases} = \left[\begin{array}{cccc|c} 3 & 2 & -1 & 1 & -1 \\ 2 & 0 & -1 & 2 & 0 \\ 3 & 1 & 2 & 5 & 2 \end{array}\right]$$

Elementary Operations

Definition There are three types of **elementary row operations** that can be performed on a matrix:

- **Row swapping**: Interchanging two rows of the matrix.
- **Row scaling**: Multiplying all entries in a row by a non-zero constant.
- **Row addition**: Adding a multiple of one row to another row.

These operations are used in methods such as Gaussian elimination to solve systems of linear equations. Row operations do not produce equal matricies, but they do produce **equivalent** matrices, meaning they represent systems of equations with the same solution set. In order for 2 matricies to be equal, they must have the same dimensions and identical entries in corresponding positions.

Example Consider the matricies:

$$\begin{bmatrix} 1 & 2 & | & 0 \\ 3 & 4 & | & 1 \end{bmatrix} \mapsto \begin{bmatrix} 2 & 4 & | & 0 \\ 3 & 4 & | & 1 \end{bmatrix}$$

This represents the row operation of scaling the first row by 2 ($2R_1 \mapsto R_1$). These matricies are equivalent but not equal.

Example Solve:

$$\begin{cases} 2x + 3y - z = 3 \\ -x - y + z = 1 \\ x - 2y - 5z = 0 \end{cases} = \begin{bmatrix} 2 & 3 & -1 & | & 3 \\ -1 & -1 & 1 & | & 1 \\ 1 & -2 & -5 & | & 0 \end{bmatrix} \xmapsto{R3 \mapsto R1} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ -1 & -1 & 1 & | & 1 \\ 2 & 3 & -1 & | & 3 \end{bmatrix} \xmapsto{R2+R1 \mapsto R2} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & -3 & -4 & | & 1 \\ 2 & 3 & -1 & | & 3 \end{bmatrix}$$

$$\xmapsto{R3-2R1 \mapsto R3} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & -3 & -4 & | & 1 \\ 0 & 7 & 9 & | & 3 \end{bmatrix} \xmapsto{-\frac{1}{3}R2 \mapsto R2} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & \frac{4}{3} & | & -\frac{1}{3} \\ 0 & 7 & 9 & | & 3 \end{bmatrix} \xmapsto{R3-7R2 \mapsto R3} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & \frac{4}{3} & | & -\frac{1}{3} \\ 0 & 0 & \frac{-1}{3} & | & \frac{16}{3} \end{bmatrix}$$

$$\xmapsto{-3R3 \mapsto R3} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & \frac{4}{3} & | & -\frac{1}{3} \\ 0 & 0 & 1 & | & -16 \end{bmatrix} \xmapsto{R2-\frac{4}{3}R3 \mapsto R2} \begin{bmatrix} 1 & -2 & -5 & | & 0 \\ 0 & 1 & 0 & | & 21 \\ 0 & 0 & 1 & | & -16 \end{bmatrix} \xmapsto{R1+5R3 \mapsto R1} \begin{bmatrix} 1 & -2 & 0 & | & -80 \\ 0 & 1 & 0 & | & 21 \\ 0 & 0 & 1 & | & -16 \end{bmatrix}$$

$$\xmapsto{R1+2R2 \mapsto R1} \begin{bmatrix} 1 & 0 & 0 & | & -38 \\ 0 & 1 & 0 & | & 21 \\ 0 & 0 & 1 & | & -16 \end{bmatrix} \implies (x, y, z) = (-38, 21, -16)$$